

TECHNICAL DOCUMENT

52Tasks

E-Commerce Operations Orchestrator

Technical Operations Manual

Architectural and Operational Specification

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Approval Register

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1. System Introduction

1.1 Document Purpose

This document constitutes the complete technical operational specification for the 52Tasks (E-Commerce Operations Orchestrator) system. Its primary purpose is to provide a detailed description of the software architecture, configuration parameters, temporal dependencies, and operational dynamics of the system, in order to support development, maintenance, testing, and operational management activities.

This document has been prepared with particular attention to knowledge transfer, thus representing a fundamental resource for onboarding new technical personnel and ensuring operational continuity of the system.

1.2 Scope of Application

The 52Tasks system represents a large-scale, complex software architecture specifically designed for the orchestration and management of a high-throughput e-commerce platform. The system was conceived to operate in a business-critical environment where reliability, performance, and efficient use of computational resources are fundamental requirements for ensuring customer satisfaction and operational success.

The system architecture is based on a distributed execution model that coordinates multiple specialized services, each responsible for specific functional domains such as payment, inventory, and logistics. These services orchestrate the execution of a large set of interdependent tasks that must comply with complex timing constraints and well-defined precedence relationships.

1.3 Intended Audience

This document is intended for: software engineers responsible for system development and maintenance, DevOps and SRE teams responsible for operational monitoring and control, quality assurance teams for verification and validation

activities, training personnel for knowledge transfer programs, and technical management for project activity supervision.

2. Global Configuration and Operational Parameters

2.1 Operational Time Window

The 52Tasks system operates within a well-defined time window extending from the initial instant, identified as time zero, up to time unit five hundred eighty-six. This extensive operational window of five hundred eighty-six time units represents the period during which all scheduled business operations, from order inception to final analytics, must be started, executed, and completed.

The choice of this specific duration is linked to the processing requirements of a complete business cycle, ensuring that all critical and background operations can be completed within the constraints imposed by service level agreements (SLAs) and data processing pipelines.

2.2 System Parameters

Parameter	Value	Description
System Name	52Tasks	Unique instance identifier
Operational Window (t_max)	586 t.u.	Maximum operational cycle duration
Max Resources (r_max)	266	Simultaneous processing units
Active Services	11	Number of service modules
Total Tasks	52	Number of operational tasks

The availability of two hundred sixty-six resources allows for a massive degree of parallelism in task execution, enabling the system to effectively manage high transaction volumes. However, this high capacity also requires sophisticated scheduling and dependency management to avoid resource contention and ensure the timely completion of critical business workflows.

3. Modular Service Architecture

3.1 Architecture Overview

The 52Tasks system architecture is structured into eleven main services, each designed to manage a specific functional domain of the e-commerce platform. This modular organization enables a clear separation of responsibilities and facilitates both maintenance and system evolution over time.

ID	Service	Primary Responsibility
1	Payment_Service	Manages all financial transaction workflows, including validation, tax, and fraud detection
2	Inventory_Service	Handles stock levels, warehouse allocation, and quality control
3	Order_Service	Orchestrates the core order lifecycle from confirmation to archival
4	Shipping_Service	Manages logistics calculations and shipping processes
5	Notification_Service	Responsible for all customer and system communications (email, SMS, etc.)
6	Security_Service	Manages compliance, data encryption, and security validations
7	Analytics_Service	Performs data analysis, reporting, and performance monitoring
8	Processing_Service	Handles generic data and media processing tasks
9	Infrastructure_Service	Manages underlying system operations like backups, caching, and health checks
10	Recommendation_Service	Manages customer-facing recommendation and loyalty systems
11	Refund_Service	Manages the dedicated refund processing workflow

3.2 Detailed Service Descriptions

3.2.1 Payment_Service (ID: 1)

The Payment_Service is central to the platform's financial operations. It manages a set of critical tasks: Payment_Validation (PAY_VAL), Tax_Calculation (TAX_CALC), Risk_Assessment (RISK_ASMT), Fraud_Detection (FRD_DET), Currency_Conversion (CUR_CONV), and Price_Optimization (PRC_OPT). This service ensures the integrity and security of all monetary transactions.

3.2.2 Inventory_Service (ID: 2)

This service is responsible for managing the physical and digital stock. It coordinates Inventory_Check (INV_CHK), Warehouse_Allocation (WH_ALLOC), Quality_Check (QC_CHK), and Return_Handling (RET_HDL). Its correct operation is fundamental for order fulfillment and supply chain management.

3.2.3 Order_Service (ID: 3)

The Order_Service orchestrates the main customer order flow. It includes Order_Confirmation (ORD_CONF), Final_Processing (FINAL_PROC), Document_Generation (DOC_GEN), Archive_Processing (ARCH_PROC), and Final_Validation (FINAL_VAL). This service acts as a central hub for the entire order lifecycle.

3.2.4 Shipping_Service (ID: 4)

A highly specialized service, the Shipping_Service is currently responsible for a single but critical task: Shipping_Calculation (SHIP_CALC). This task determines logistics costs and options, a crucial input for the checkout process.

3.2.5 Notification_Service (ID: 5)

This service manages all outgoing communications. It orchestrates Customer_Notification (CUST_NOTIF), Email_Dispatch (EMAIL_DISP), SMS_Notification (SMS_NOTIF), Push_Notification (PUSH_NOTIF), and Webhook_Trigger (HOOK_TRIG), ensuring timely and multi-channel information delivery.

3.2.6 Security_Service (ID: 6)

The Security_Service is responsible for maintaining the platform's security and compliance posture. It runs Compliance_Check (COMP_CHK), Security_Validation (SEC_VAL), Data_Encryption (DATA_ENCR), Audit_Trail (AUD_TRL), API_Validation (API_VAL), and Session_Management (SESS_MGT).

3.2.7 Analytics_Service (ID: 7)

This service provides business intelligence and operational monitoring. Its tasks include Performance_Monitor (PERF_MON), Report_Generation (RPT_GEN), Analytics_Update (ANLY_UPD), Search_Indexing (SRCH_IDX), Recommendation_Engine (REC_ENG), and AB_Testing (AB_TEST).

3.2.8 Processing_Service (ID: 8)

The Processing_Service handles various data and media processing tasks that support the platform. It includes Content_Moderation (CONT_MOD),

Image_Processing (IMG_PROC), Video_Processing (VID_PROC), File_Upload (FILE_UPL), and Metadata_Extraction (META_EXTR).

3.2.9 Infrastructure_Service (ID: 9)

This service is responsible for the health and maintenance of the underlying system infrastructure. It manages critical background tasks such as Backup_Creation (BCK_CREA), System_Cleanup (SYS_CLN), Cache_Refresh (CACHE_REFR), Log_Processing (LOG_PROC), Error_Handling (ERR_HDL), Load_Balancing (LOAD_BAL), Database_Sync (DB_SYNC), and Health_Check (HLTH_CHK).

3.2.10 Recommendation_Service (ID: 10)

Focused on enhancing user experience and engagement, this service manages Loyalty_Processing (LOY_PROC), Product_Recommendation (PRD_REC), Feature_Toggle (FEAT_TGL), Rate_Limiting (RATE_LMT), and Circuit_Breaker (CIRC_BRK) tasks.

3.2.11 Refund_Service (ID: 11)

A dedicated service for handling post-purchase financial adjustments, it contains the single task Refund_Processing (REF_PROC). This ensures that returns and refunds are handled through a specialized and isolated workflow.

4. Detailed Description of Operational Tasks

4.1 Complete Task Registry

ID	Task Code	Service Name	Duration	Resources	Max Concur.	Rep
1	PAY_VAL	Payment_Service	5	4	4	0/0
2	INV_CHK	Inventory_Service	4	5	5	3/0
3	SHIP_CALC	Shipping_Service	10	3	3	3/0
4	ORD_CONF	Order_Service	8	2	2	0/0
5	CUST_NOTIF	Notification_Service	8	5	1	0/0
6	WH_ALLOC	Inventory_Service	9	4	1	3/16
7	QC_CHK	Inventory_Service	6	4	2	1/0

ID	Task Code	Service Name	Duration	Resources	Max Concur.	Rep
8	FINAL_PROC	Order_Service	5	3	5	0/0
9	TAX_CALC	Payment_Service	3	5	1	2/0
10	DOC_GEN	Order_Service	9	5	2	2/4
11	COMP_CHK	Security_Service	10	5	4	0/0
12	ARCH_PROC	Order_Service	6	5	2	3/16
13	SEC_VAL	Security_Service	6	1	3	1/0
14	PERF_MON	Analytics_Service	4	1	3	2/0
15	BCK_CREA	Infrastructure_Service	8	2	5	1/0
16	RPT_GEN	Analytics_Service	3	3	5	0/0
17	DATA_ENCR	Security_Service	10	5	1	0/0
18	AUD_TRL	Security_Service	6	2	1	1/0
19	SYS_CLN	Infrastructure_Service	4	1	3	0/4
20	FINAL_VAL	Order_Service	3	3	4	0/0
21	RISK_ASMT	Payment_Service	10	1	5	1/0
22	FRD_DET	Payment_Service	6	2	5	3/8
23	CUR_CONV	Payment_Service	10	5	4	0/0
24	PRC_OPT	Payment_Service	3	3	3	0/0
25	LOY_PROC	Recommendation_Service	6	5	4	1/0
26	RET_HDL	Inventory_Service	3	1	3	2/8
27	REF_PROC	Refund_Service	4	1	2	1/12
28	PRD_REC	Recommendation_Service	9	5	4	0/0
29	ANLY_UPD	Analytics_Service	4	2	5	0/14
30	CACHE_REFR	Infrastructure_Service	3	1	3	0/9
31	LOG_PROC	Infrastructure_Service	10	4	3	0/0
32	ERR_HDL	Infrastructure_Service	8	3	5	1/0
33	LOAD_BAL	Infrastructure_Service	7	1	1	1/0
34	DB_SYNC	Infrastructure_Service	7	2	4	0/0
35	API_VAL	Security_Service	5	1	4	2/7
36	SESS_MGT	Security_Service	6	5	3	0/0

ID	Task Code	Service Name	Duration	Resources	Max Concur.	Rep
37	EMAIL_DISP	Notification_Service	8	4	2	3/0
38	SMS_NOTIF	Notification_Service	5	5	2	3/6
39	PUSH_NOTIF	Notification_Service	7	5	2	0/0
40	HOOK_TRIG	Notification_Service	5	3	2	0/3
41	CONT_MOD	Processing_Service	8	2	4	0/0
42	IMG_PROC	Processing_Service	9	3	2	2/8
43	VID_PROC	Processing_Service	6	1	1	0/0
44	FILE_UPL	Processing_Service	7	5	5	3/7
45	META_EXTR	Processing_Service	6	2	1	0/0
46	SRCH_IDX	Analytics_Service	10	2	3	3/15
47	REC_ENG	Analytics_Service	5	4	2	2/0
48	AB_TEST	Analytics_Service	5	2	4	0/0
49	FEAT_TGL	Recommendation_Service	3	3	2	2/0
50	RATE_LMT	Recommendation_Service	6	1	5	0/0
51	CIRC_BRK	Recommendation_Service	10	3	4	2/0
52	HLTH_CHK	Infrastructure_Service	9	1	3	3/12

Legend: Duration = time units; Resources = required computational units; Max Concur. = maximum concurrent executions; Rep = repetitions (rc/rd format where rc = repeat count, rd = repeat delay)

The system is characterized by a high degree of heterogeneity in task specifications. Resource requirements (`res_q`) range from 1 to 5 units, and maximum concurrency (`max_c`) varies significantly. Many tasks are repetitive, with diverse repeat counts and delays, reflecting continuous monitoring, periodic batch jobs, and recurring checks.

5. Temporal Constraints and Dependency System

5.1 Dual-Mode Start-to-Start Precedence Model

The 52Tasks system implements a sophisticated temporal constraint system based on the concept of start-to-start precedence. A destination task can begin execution only after a given delay has elapsed since the start of the source task execution.

This system employs a dual-mode mechanism controlled by the `wait_all` flag, which dictates the behavior for repetitive source tasks:

- * `wait_all: false`: The destination task can start after the specified delay has passed from the beginning of the *first* execution of the source task. This allows for rapid, overlapping execution and high parallelism.
- * `wait_all: true`: The destination task must wait for *all* repetitions of the source task to complete. The delay timer begins only after the final repetition has started. This enforces a stricter synchronization, ensuring a complete cycle of the source task is finished before its dependent proceeds.

This dual-mode model provides fine-grained control over the operational flow, balancing the need for speed with the requirement for sequential integrity.

5.2 Dependency Matrix

Source Task	Destination Task	Delay (t.u.)	Type
Payment_Validation (1)	Inventory_Check (2)	5	SS (wait_all=false)
Inventory_Check (2)	Shipping_Calculation (3)	9	SS (wait_all=true)
Payment_Validation (1)	Order_Confirmation (4)	21	SS (wait_all=false)
Inventory_Check (2)	Order_Confirmation (4)	16	SS (wait_all=true)
Shipping_Calculation (3)	Order_Confirmation (4)	10	SS (wait_all=true)
Payment_Validation (1)	Customer_Notification (5)	6	SS (wait_all=false)
Shipping_Calculation (3)	Customer_Notification (5)	12	SS (wait_all=true)
Customer_Notification (5)	Warehouse_Allocation (6)	13	SS (wait_all=false)
Payment_Validation (1)	Quality_Check (7)	13	SS (wait_all=true)
Customer_Notification (5)	Quality_Check (7)	8	SS (wait_all=false)
Order_Confirmation (4)	Quality_Check (7)	12	SS (wait_all=false)

Source Task	Destination Task	Delay (t.u.)	Type
Quality_Check (7)	Final_Processing (8)	12	SS (wait_all=true)
Shipping_Calculation (3)	Final_Processing (8)	11	SS (wait_all=false)
Payment_Validation (1)	Tax_Calculation (9)	7	SS (wait_all=false)
Warehouse_Allocation (6)	Tax_Calculation (9)	15	SS (wait_all=false)
Customer_Notification (5)	Tax_Calculation (9)	6	SS (wait_all=true)
Tax_Calculation (9)	Document_Generation (10)	7	SS (wait_all=false)
Shipping_Calculation (3)	Compliance_Check (11)	12	SS (wait_all=true)
Order_Confirmation (4)	Compliance_Check (11)	12	SS (wait_all=true)
Warehouse_Allocation (6)	Archive_Processing (12)	16	SS (wait_all=true)
Tax_Calculation (9)	Archive_Processing (12)	6	SS (wait_all=true)
Document_Generation (10)	Security_Validation (13)	16	SS (wait_all=false)
Customer_Notification (5)	Security_Validation (13)	16	SS (wait_all=true)
Tax_Calculation (9)	Security_Validation (13)	11	SS (wait_all=false)
Warehouse_Allocation (6)	Performance_Monitor (14)	14	SS (wait_all=true)
Shipping_Calculation (3)	Backup_Creation (15)	16	SS (wait_all=true)
Backup_Creation (15)	Report_Generation (16)	15	SS (wait_all=true)
Document_Generation (10)	Report_Generation (16)	6	SS (wait_all=true)
Security_Validation (13)	Data_Encryption (17)	5	SS (wait_all=false)
Payment_Validation (1)	Audit_Trail (18)	8	SS (wait_all=true)
Report_Generation (16)	Audit_Trail (18)	14	SS (wait_all=false)
Archive_Processing (12)	System_Cleanup (19)	14	SS (wait_all=true)
Backup_Creation (15)	System_Cleanup (19)	8	SS (wait_all=false)
Backup_Creation (15)	Final_Validation (20)	7	SS (wait_all=true)
Archive_Processing (12)	Final_Validation (20)	15	SS (wait_all=false)
Security_Validation (13)	Final_Validation (20)	5	SS (wait_all=true)
Data_Encryption (17)	Risk_Assessment (21)	10	SS (wait_all=true)
Report_Generation (16)	Fraud_Detection (22)	7	SS (wait_all=false)
Data_Encryption (17)	Currency_Conversion (23)	14	SS (wait_all=false)
Audit_Trail (18)	Currency_Conversion (23)	5	SS (wait_all=true)

Source Task	Destination Task	Delay (t.u.)	Type
Final_Validation (20)	Currency_Conversion (23)	9	SS (wait_all=false)
Shipping_Calculation (3)	Price_Optimization (24)	13	SS (wait_all=true)
Performance_Monitor (14)	Price_Optimization (24)	9	SS (wait_all=false)
Risk_Assessment (21)	Loyalty_Processing (25)	10	SS (wait_all=false)
Risk_Assessment (21)	Return_Handling (26)	6	SS (wait_all=true)
Fraud_Detection (22)	Return_Handling (26)	12	SS (wait_all=false)
Audit_Trail (18)	Return_Handling (26)	11	SS (wait_all=false)
Price_Optimization (24)	Refund_Processing (27)	14	SS (wait_all=false)
Final_Validation (20)	Refund_Processing (27)	13	SS (wait_all=false)
Final_Validation (20)	Product_Recommendation (28)	16	SS (wait_all=false)
Final_Processing (8)	Product_Recommendation (28)	8	SS (wait_all=true)
Fraud_Detection (22)	Analytics_Update (29)	16	SS (wait_all=true)
Performance_Monitor (14)	Cache_Refresh (30)	10	SS (wait_all=true)
Inventory_Check (2)	Cache_Refresh (30)	5	SS (wait_all=true)
Compliance_Check (11)	Cache_Refresh (30)	16	SS (wait_all=false)
Document_Generation (10)	Log_Processing (31)	13	SS (wait_all=false)
Data_Encryption (17)	Log_Processing (31)	5	SS (wait_all=true)
Cache_Refresh (30)	Error_Handling (32)	14	SS (wait_all=true)
Analytics_Update (29)	Error_Handling (32)	7	SS (wait_all=true)
Product_Recommendation (28)	Error_Handling (32)	10	SS (wait_all=false)
Error_Handling (32)	Load_Balancing (33)	15	SS (wait_all=true)
Log_Processing (31)	Load_Balancing (33)	12	SS (wait_all=true)
Loyalty_Processing (25)	Load_Balancing (33)	15	SS (wait_all=false)
Cache_Refresh (30)	Database_Sync (34)	13	SS (wait_all=false)
Refund_Processing (27)	Database_Sync (34)	14	SS (wait_all=false)
Order_Confirmation (4)	Database_Sync (34)	13	SS (wait_all=true)
Error_Handling (32)	API_Validation (35)	10	SS (wait_all=true)
Product_Recommendation (28)	Session_Management (36)	10	SS (wait_all=false)
Log_Processing (31)	Email_Dispatch (37)	11	SS (wait_all=true)

Source Task	Destination Task	Delay (t.u.)	Type
Load_Balancing (33)	Email_Dispatch (37)	14	SS (wait_all=true)
API_Validation (35)	Email_Dispatch (37)	9	SS (wait_all=false)
Session_Management (36)	SMS_Notification (38)	8	SS (wait_all=false)
Cache_Refresh (30)	SMS_Notification (38)	12	SS (wait_all=false)
Load_Balancing (33)	SMS_Notification (38)	8	SS (wait_all=false)
Log_Processing (31)	Push_Notification (39)	8	SS (wait_all=true)
Push_Notification (39)	Webhook_Trigger (40)	7	SS (wait_all=true)
Session_Management (36)	Webhook_Trigger (40)	10	SS (wait_all=false)
Final_Validation (20)	Content_Moderation (41)	6	SS (wait_all=true)
Risk_Assessment (21)	Content_Moderation (41)	8	SS (wait_all=true)
Document_Generation (10)	Content_Moderation (41)	12	SS (wait_all=false)
Final_Processing (8)	Image_Processing (42)	5	SS (wait_all=true)
Payment_Validation (1)	Image_Processing (42)	10	SS (wait_all=false)
Final_Validation (20)	Image_Processing (42)	5	SS (wait_all=true)
Final_Validation (20)	Video_Processing (43)	9	SS (wait_all=true)
Inventory_Check (2)	Video_Processing (43)	10	SS (wait_all=true)
Email_Dispatch (37)	Video_Processing (43)	8	SS (wait_all=false)
Video_Processing (43)	File_Upload (44)	6	SS (wait_all=false)
Content_Moderation (41)	File_Upload (44)	7	SS (wait_all=true)
SMS_Notification (38)	Metadata_Extraction (45)	13	SS (wait_all=false)
File_Upload (44)	Metadata_Extraction (45)	13	SS (wait_all=false)
Image_Processing (42)	Search_Indexing (46)	8	SS (wait_all=true)
Metadata_Extraction (45)	Search_Indexing (46)	6	SS (wait_all=false)
API_Validation (35)	Recommendation_Engine (47)	7	SS (wait_all=true)
Analytics_Update (29)	Recommendation_Engine (47)	12	SS (wait_all=true)
Order_Confirmation (4)	Recommendation_Engine (47)	6	SS (wait_all=false)
Search_Indexing (46)	AB_Testing (48)	7	SS (wait_all=false)
Search_Indexing (46)	Feature_Toggle (49)	16	SS (wait_all=true)
Recommendation_Engine (47)	Feature_Toggle (49)	12	SS (wait_all=true)

Source Task	Destination Task	Delay (t.u.)	Type
Image_Processing (42)	Feature_Toggle (49)	6	SS (wait_all=false)
Feature_Toggle (49)	Rate_Limiting (50)	12	SS (wait_all=true)
Recommendation_Engine (47)	Circuit_Breaker (51)	10	SS (wait_all=false)
Feature_Toggle (49)	Health_Check (52)	10	SS (wait_all=false)

Legend: SS = Start-to-Start

5.3 Critical Dependency Analysis

The dependency graph of the 52Tasks system is extensive and highly interconnected, reflecting a tightly coupled business process. The initial flow is driven primarily by `Payment_Validation` (1) and `Inventory_Check` (2), which trigger numerous parallel and sequential paths.

A key convergence point is `Order_Confirmation` (4), which depends on the completion of `Payment_Validation`, `Inventory_Check`, and `Shipping_Calculation`. The `wait_all=true` constraints from the latter two ensure that all their repetitions (representing, for example, multiple stock checks or shipping rate lookups) are complete before the order is finalized, guaranteeing data consistency.

The system features numerous "fan-out" points where a single task triggers multiple downstream processes. For example, `Final_Validation` (20) initiates `Currency_Conversion` (23), `Refund_Processing` (27), `Product_Recommendation` (28), `Content_Moderation` (41), `Image_Processing` (42), and `Video_Processing` (43), showcasing a major branching of workflows after final order validation.

The strategic use of `wait_all=true` versus `wait_all=false` is critical. For instance, `Inventory_Check` (2) to `Shipping_Calculation` (3) uses `wait_all=true`, forcing shipping to wait for all inventory checks. Conversely, `Payment_Validation` (1) to `Customer_Notification` (5) uses `wait_all=false`, allowing notifications to be sent quickly after the initial validation without waiting for subsequent checks. This design choice carefully balances process integrity with system responsiveness.

6. Final Considerations and Recommendations

6.1 Architecture Summary

The 52Tasks system represents a powerful and comprehensive architecture for orchestrating complex, large-scale e-commerce operations. Its design reflects a mature understanding of business process requirements, high-throughput data pipelines, and the need for robust, parallel execution.

The balance between a large number of processing resources and a highly complex web of temporal dependencies demonstrates a sophisticated approach to critical system design. The heterogeneity of tasks in terms of duration, resource consumption, and repetition patterns highlights the system's flexibility in handling a wide variety of operational workloads.

6.2 Modular Architecture Benefits

The fine-grained modularity of the architecture, with its clear separation into eleven specialized services, is essential for managing the system's complexity. This structure facilitates system comprehension, targeted maintenance, and future evolution. The ability to modify or extend individual services (e.g., adding a new payment method in `Payment_Service`) without impacting the entire architecture is a significant advantage for adapting to evolving business needs.

The dual-mode precedence model (`wait_all` flag) provides the necessary control to build complex and reliable workflows, allowing schedulers to optimize for either speed or consistency depending on the specific dependency.

6.3 Knowledge Transfer Recommendations

For effective knowledge transfer, it is recommended to follow a structured training path that includes: understanding global parameters and the extensive operational window, studying the service architecture and the responsibility of each of the eleven services, detailed analysis of task characteristics with emphasis on their diverse resource and repetition profiles, mapping the complex temporal dependencies and mastering the dual-mode start-to-start model (`wait_all` semantics), simulating critical workflows such as the main order path (`Payment_Validation` -> `Inventory_Check` -> `Order_Confirmation`), and finally,

studying the role of infrastructure and security services in ensuring overall system stability and integrity.

Appendix A: Glossary

Term	Definition
Orchestrator	A system designed to manage and coordinate complex workflows and services.
t.u.	Time Unit - Base unit for time measurement in the system.
r_max	Maximum number of computational resources allocable simultaneously.
t_max	Maximum duration of operational window in time units.
Start-to-Start (SS)	Type of precedence constraint based on task start rather than completion.
wait_all	A constraint flag that dictates synchronization behavior. <code>false</code> allows the dependent task to start after a delay from the <i>first</i> execution of a repetitive source task. <code>true</code> requires <i>all</i> repetitions of the source task to complete before the delay is applied.
Repetitive Task	A task configured to execute multiple times at defined intervals.
rc	Repeat Count - Number of repetitions for a repetitive task.
rd	Repeat Delay - Time delay between repetitions of a repetitive task.
max_c	Maximum Concurrency - Maximum number of concurrent executions allowed for a single task type.
Pipeline	A processing sequence where the output of one task or service becomes the input of the next.

Appendix B: Reference Documents

Code	Document Title	Version
REF-001	52Tasks System Requirements Specification	[To be filled]
REF-002	Interface Control Document - Microservices	[To be filled]
REF-003	Verification and Validation Plan	[To be filled]

Code	Document Title	Version
REF-004	E-Commerce Platform Operations Manual	[To be filled]

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